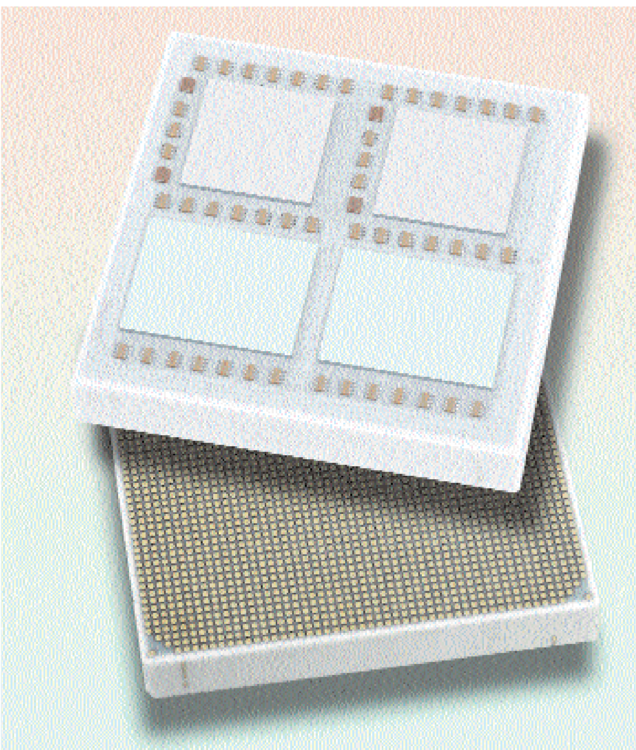
might want to upgrade to a good-quality display panel soon, because Vista will demand it (we know you'll upgrade anyway). And third, take comfort in that essential fact of human nature: you get used to what you have.

PROCESSORS

Multi-core is the new gigahertz, at least in war-speak. Dual-core was a fancy term even in mid-2005; in 2006, dual-core processors became so cheap, it made no sense to go with a single-core any more. Dual-cores, once thought of as hype, turned out to give significant performance gains when multi-tasking, and even with single tasks like video encoding.



Four cores on a chip: very welcome indeed!

And why stop there, once the first power of two had been breached? Intel's quad-core Xeon 5300 and Core 2 Extreme QX6700 were unveiled in November, making it to the second power much before AMD. Some software—such as Adobe Premiere Pro—can already take advantage of four cores, but as you'd expect, there are far fewer such programs than can put two cores to good use. The QX6700, however, in tests, showed it could spread the workload when handed two or three processor-intensive tasks. Enthusiasts—those of the deep-pocketed persuasion—were happy.

The CPU and GPU are engaged in some major billing and cooing

AMD will reveal their own quad-cores—currently called "Barcelona"—sometime in mid-2007, so there's no reason to get excited yet. But anyway, AMD is never to be left behind, as the phrase goes, and has figured out how to offer bleeding-edge performance by placing two Athlon 64 dual-core chips in one machine: the "4x4", which will probably have begun shipping by the time you read this.

The big surprise merger of the year (apart from Google and YouTube) for us was, of course, AMD taking over ATI. The effects? Delightful twists in the AMD/Intel saga, of course. "Fusion" technology is already in the works: this new kind of processor integrates CPU and GPU at the silicon level, which could mean... a lot of things. Better frame rates, anyone?

Perhaps more than that. AMD indicated that it is developing more than just one CPU/GPU platform, which will make for "step-function increases in performance-per-watt relative to today's CPU-only architectures." In press releases, AMD didn't describe the new chips as high-end graphics solutions, but as solutions that provide the "best customer experience." Fusion is expected to debut in late 2008—quite a while off, so no point getting excited, again.

But they're already talking about "Stream computing" for now. From the new ati.amd.com: "Introducing the AMD Stream Processor—the first, dedicated, GPU-based solution to address the needs of high-performance computing users. Featuring the AMD R580 GPU, and a 1GB memory buffer... the AMD Stream Processor... (enables) the most direct access to GPU resources in the industry."

Yes, the CPU and GPU are engaged in some major billing and cooing. GPUs are taking on a more generic role—you can already offload calculations onto the GPU, which could traditionally only be done on the CPU. For example, Stanford University has announced new software that will enable the use of *graphics* cards—not just CPUs—in their Folding@Home distributed computing project. (See *We Want You, Digit,* November 2006.)

Breaking away from the powers-of-two rule, Intel is talking about an 80-core CPU. They call it a "terascale research chip": 80 cores on one piece of silicon. Who'd need that and why is a question for another day, another year. Will the GPU become a core—or a few—on the CPU? Or the other way round? Who's to tell?